

GPS Project Planning, Data Collection, Map Projections & Coordinate System



Objectives

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- Know the concepts of map projection and coordinate system
- Know the basic principles of map projections, and the details of some commonly used projections
- Know the process of GPS Project Planning

Georeferencing

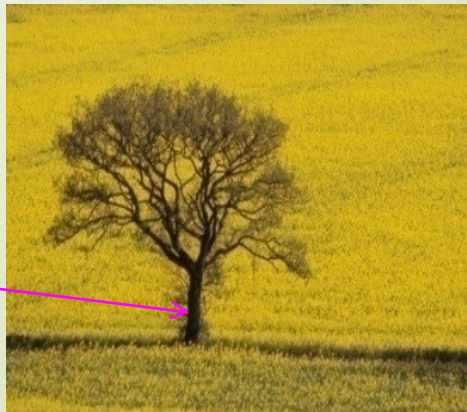
Spatial vs. aspatial: Spatial data means that the data are georeferenced, data are said to be aspatial or non-spatial with no geographic information.

Georeferencing: also called geolocate or geocode, means assigning location to data and information.

Principles for georeferencing:

- Georeference should be **unique**, that is, only one location associate with a given georeference.
- Georeferences should stay **constant** through time.
- **Metric** georeferences are much more useful, because they allow maps to be made and distances to be calculated.

(x, y)

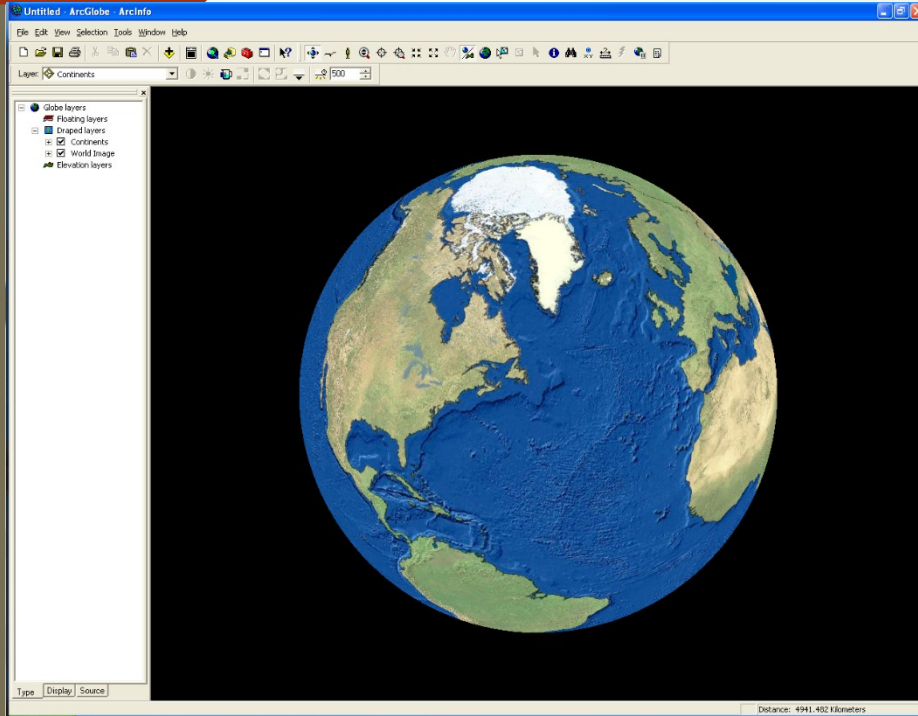


Georeferencing

Commonly used systems of georeferencing

1. **Place names** – i.e., London, ON, Canada
2. **Postal addresses and postal codes** – work well for georeferencing houses and offices, but not natural features
3. The **US public land survey system (PLSS)** – Western USA only, unique to prime meridian, example: Sec 5, Township 4N, Range 6E
4. **Latitude/longitude** – global, example: 119⁰45'W, 34⁰40'N
5. **Universal Transverse Mercator** – zones (6 degrees of longitude wide, N and S hemisphere, example: 566765E, 4356732N.

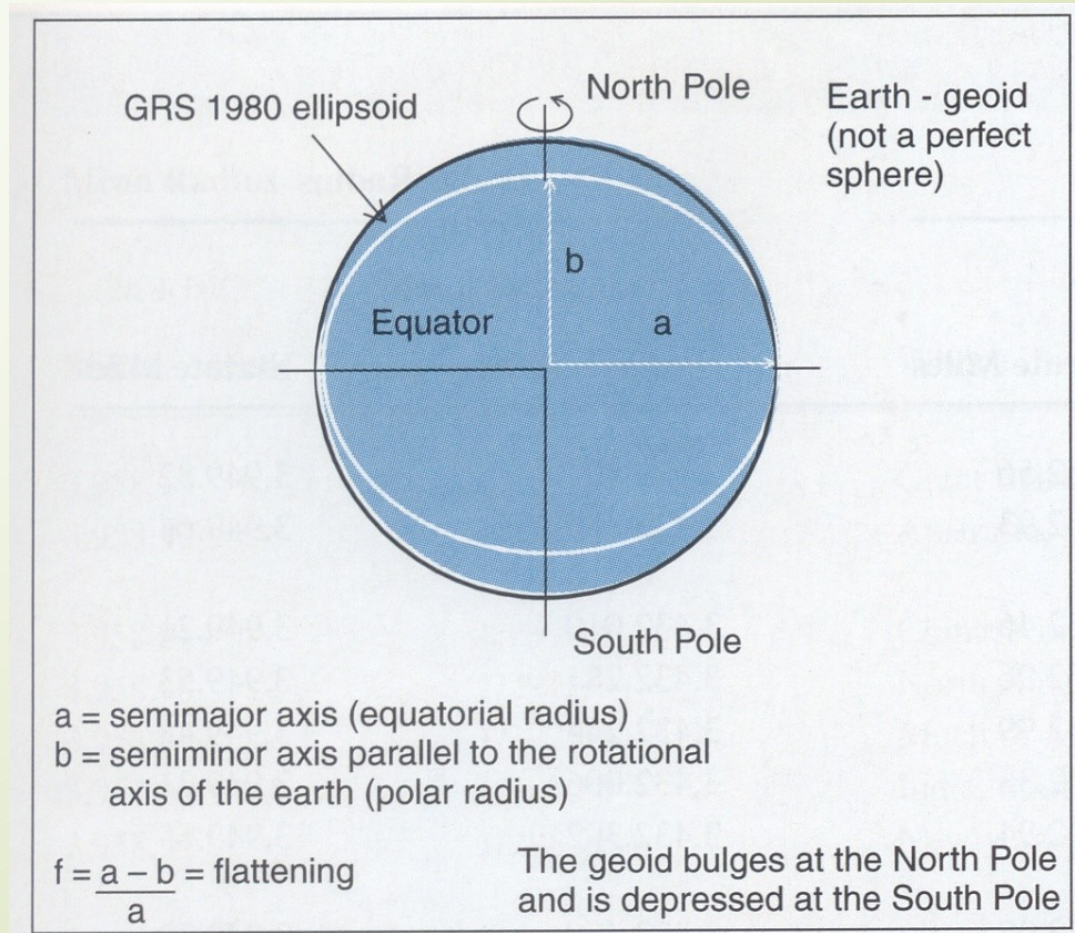
From Earth to flat map



- Flat maps are convenient.
- Round earth must be transformed to flat maps.
- Based on plane and spherical geometry.

Size and shape of the earth

- **Shape of the earth:**
Flattened at the poles and extended at the equator.



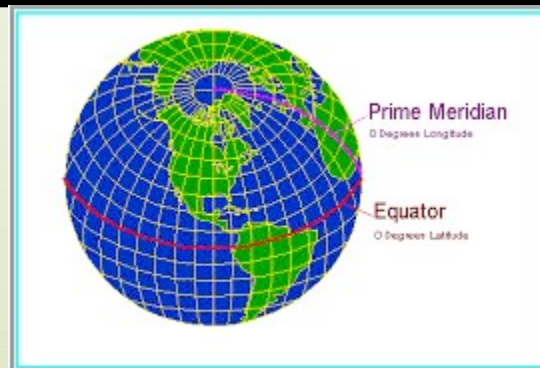
Earth coordinate systems or geometry

The locations of any objects can be precisely determined based on coordinate systems developed based on earth and plane coordinate geometry .

Earth coordinate system

- Longitude, latitude
- Decimal degree
- Degrees, minutes, seconds

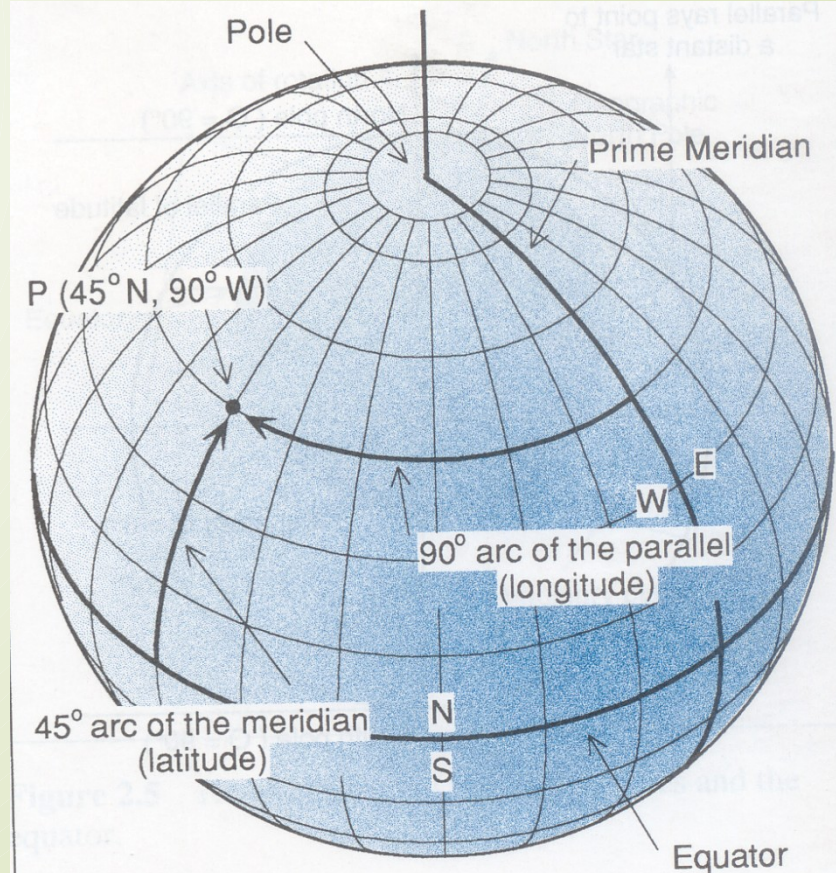
Determining locations on the earth is based on sexagesimal ($1^{\circ}=60'$) scale. That is, a circle is 360 degrees, each degree into 60 minutes, and each minute into 60 seconds.



Earth coordinate geometry

Earth coordinate geometry: The system consisting of latitude parallels and meridians is called geographic grid and means the earth spherical coordinate geometry.

- 1. Latitude:** No parallel is equal in length to any other. Parallels decrease in length at high latitudes.
- 2. Longitude:** All meridians are of equal length and one-half the perimeter. But, the polar flattening affects the lengths of meridians.
- 3. Graticule:** is a combination of latitude and longitude.



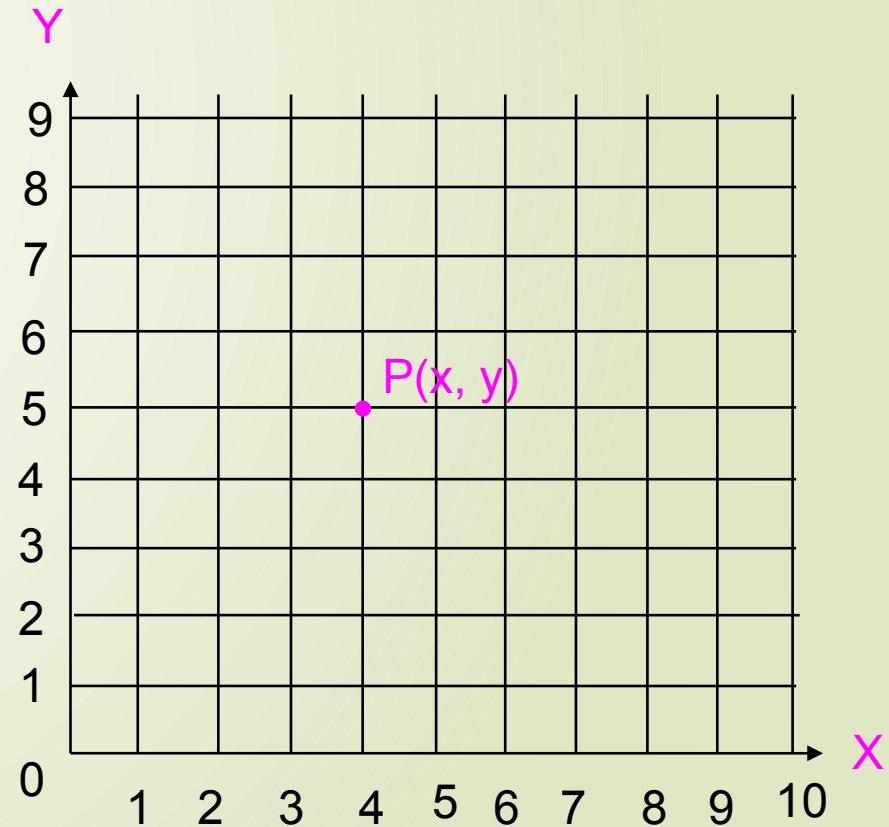
Plane coordinate geometry

Plane coordinate geometry:

Cartesian coordinate system consists of the horizontal axis referred to the x-axis and the vertical axis to the y-axis and defines the location of any point from the origin.

The plane of Cartesian space is marked at intervals by equally space lines. Each point has its own unique location with the values of x-axis and y-axis. For example, point p (x, y) is (4, 5)

- Cartesian space (x, y)
- Decimal degree
- Miles, meters, etc.

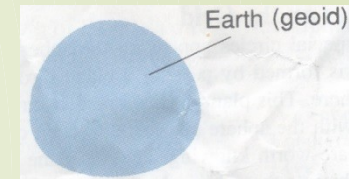


Map projection

Map projection process: Earth → flat map

- Selecting reference ellipsoid: choosing the semimajor and semi minor axes and obtaining an ellipsoids that best describes the real earth.
- Selecting developable surface: Reference ellipsoid developable surface.
- Transformation: transforming the earth to flat map, that is, a map projection is generated.

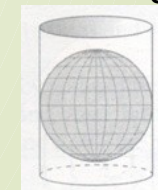
Map projection: is the transformation of the spherical surface to a plane surface , that is, transforming a position on the Earth surface identified by latitude and longitude (ϕ, λ) into a position in Cartesian coordinates (x, y).



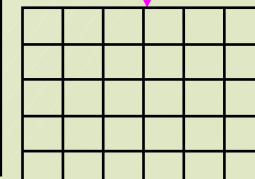
Best fit by geodesy



Select developable surface



Transformation



Map projection



Surface transformation

Four kinds of surface transformations:

1. Equal-area projection
2. Conformal projection
3. Equidistance projection
4. Azimuthal projection

Principle: select the projections that minimize errors.

Surface transformation

Equal-Area Projection (or equivalent projection):

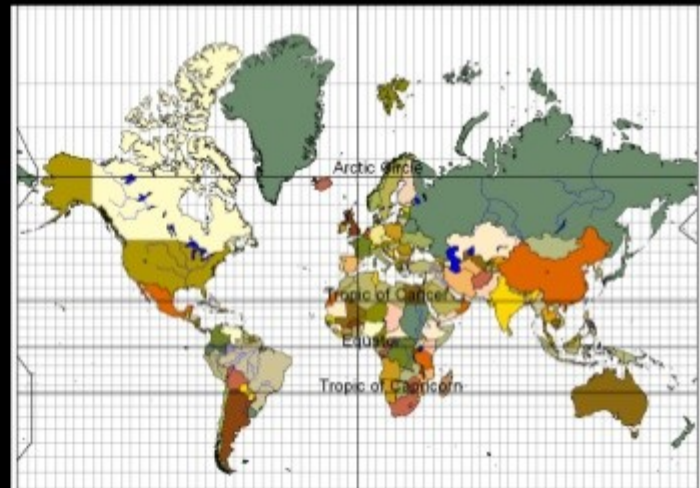
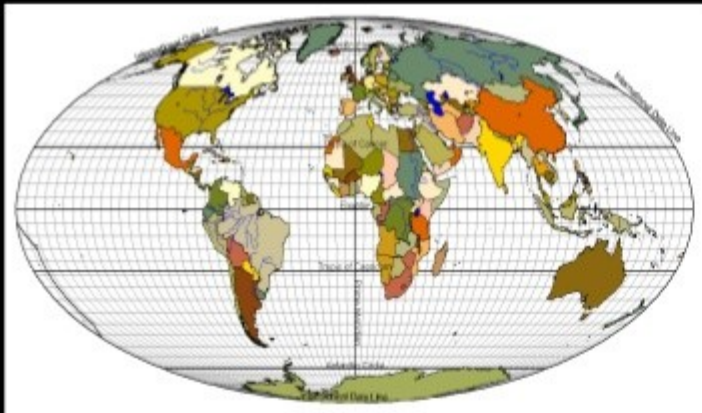
- No area deformation
- Linear or distance distortions often take place.
- Shape is skewed. Therefore, the intersections of meridians and parallels are not at right angles.
- Used to maintain correct area properties and calculate areas.



Surface transformation

Conformal Projection (or orthomorphic projection)

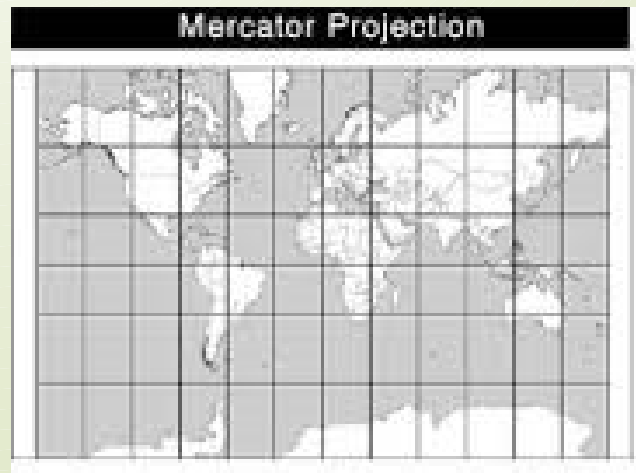
- Preserve angles at points.
- Preserve shapes of small areas, but may severely distort shapes of large areas. Areas are also distorted. For large-scale mapping of small areas, distortion is not significant.
- Meridians intersect parallels at right angles.
- The scale is the same in all directions about a point.
- Useful for navigation



Surface transformation

Azimuthal projection:

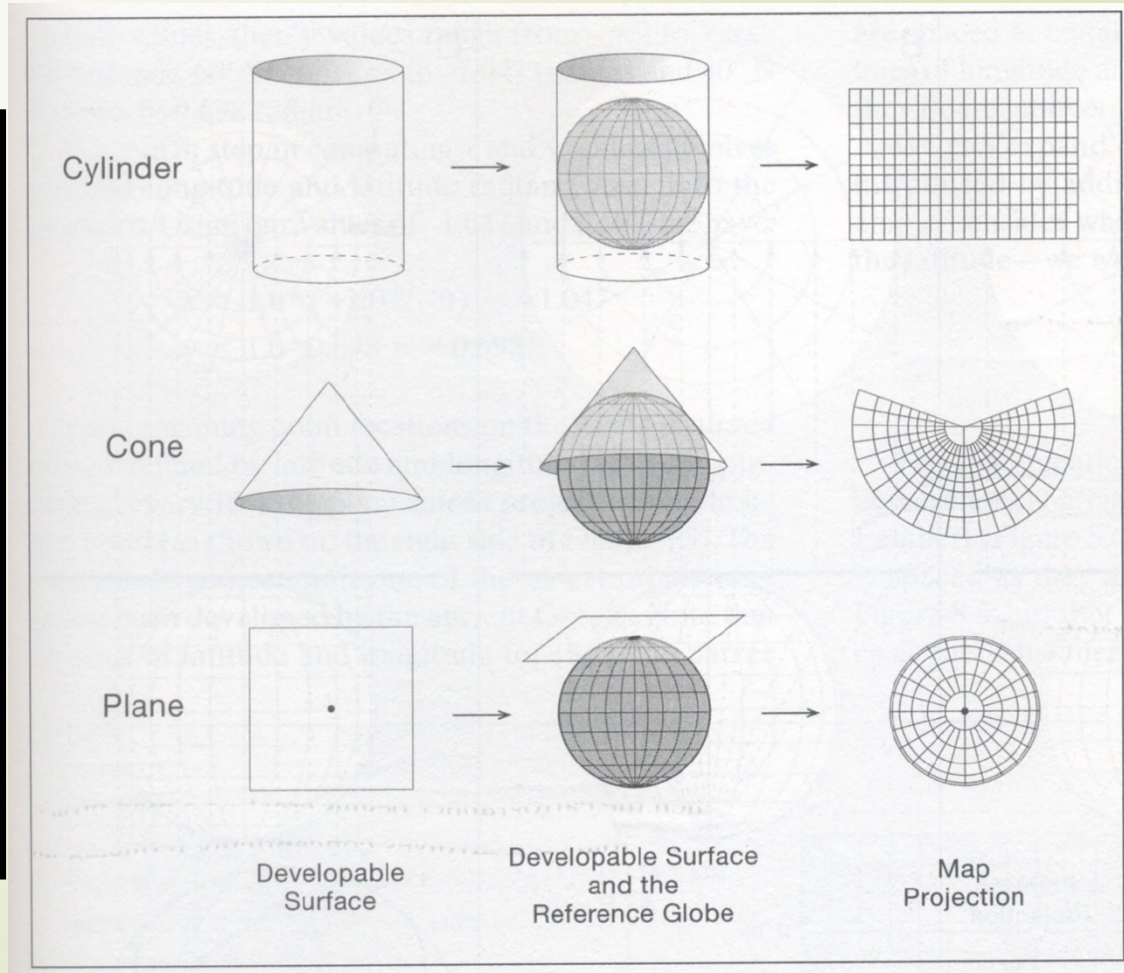
- Directions from the projection's center to all points are correct.
- Directions from points other than the center to other points are not accurate.



Map projections

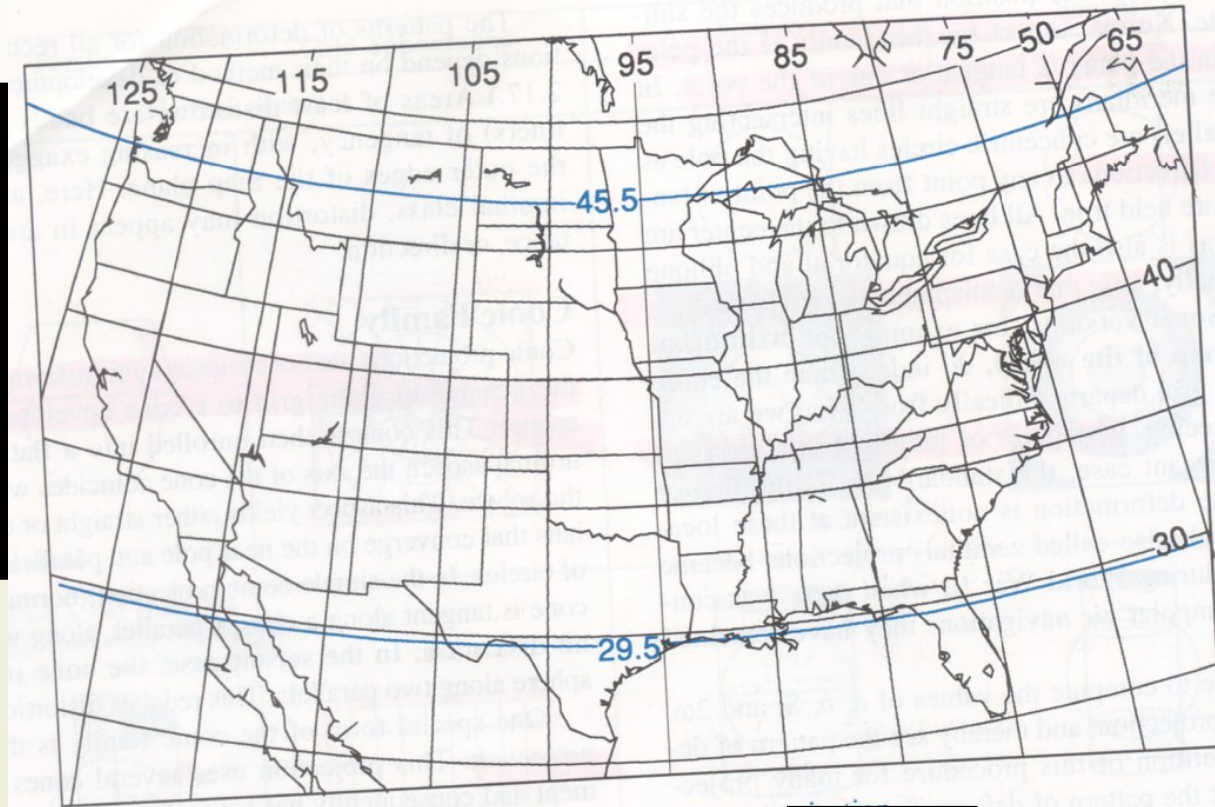
Three basic forms of developable surface: plane, cylinder, and cone, which leads to three families of map projections:

1. Cylindrical projection
2. Conic projection
3. Planar or azimuthal projection



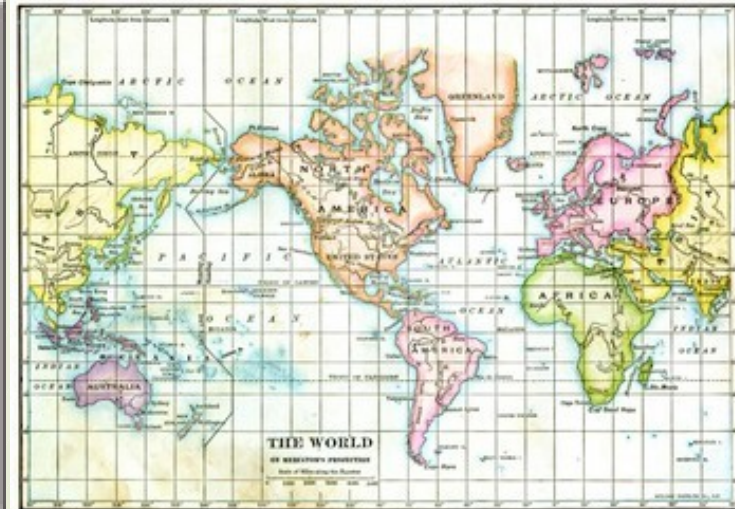
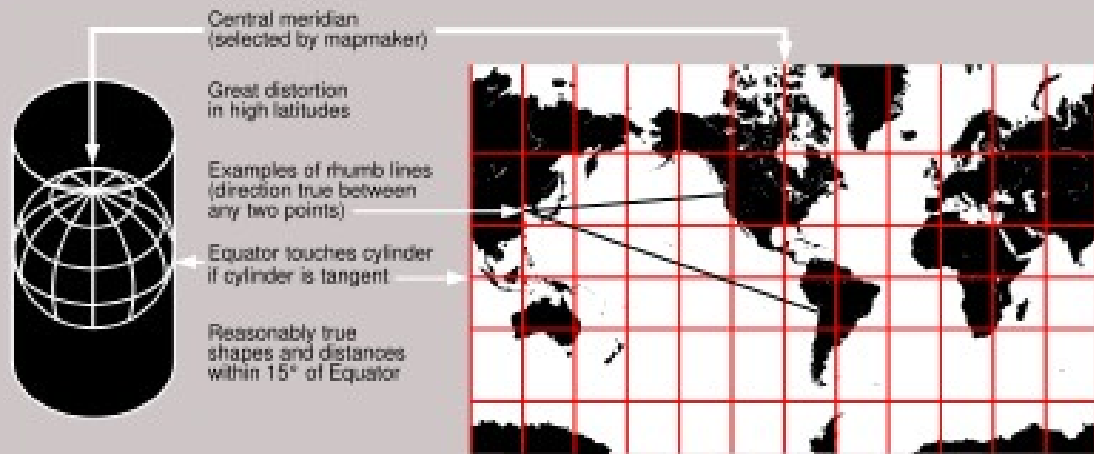
Conic projection: examples

Albers conic equal-area projection, with standard parallels at $29\frac{1}{2}$ degrees and $45\frac{1}{2}$ degree, is good choice for mapping the United States.



Dent
1999

Mercator Projection



Mercator Projection: Parallel spacing is calculated to maintain conformality. Not equal-area, or equidistant. It is useful for navigation, however, the extreme area distortion makes it unsuitable for general maps of large areas.

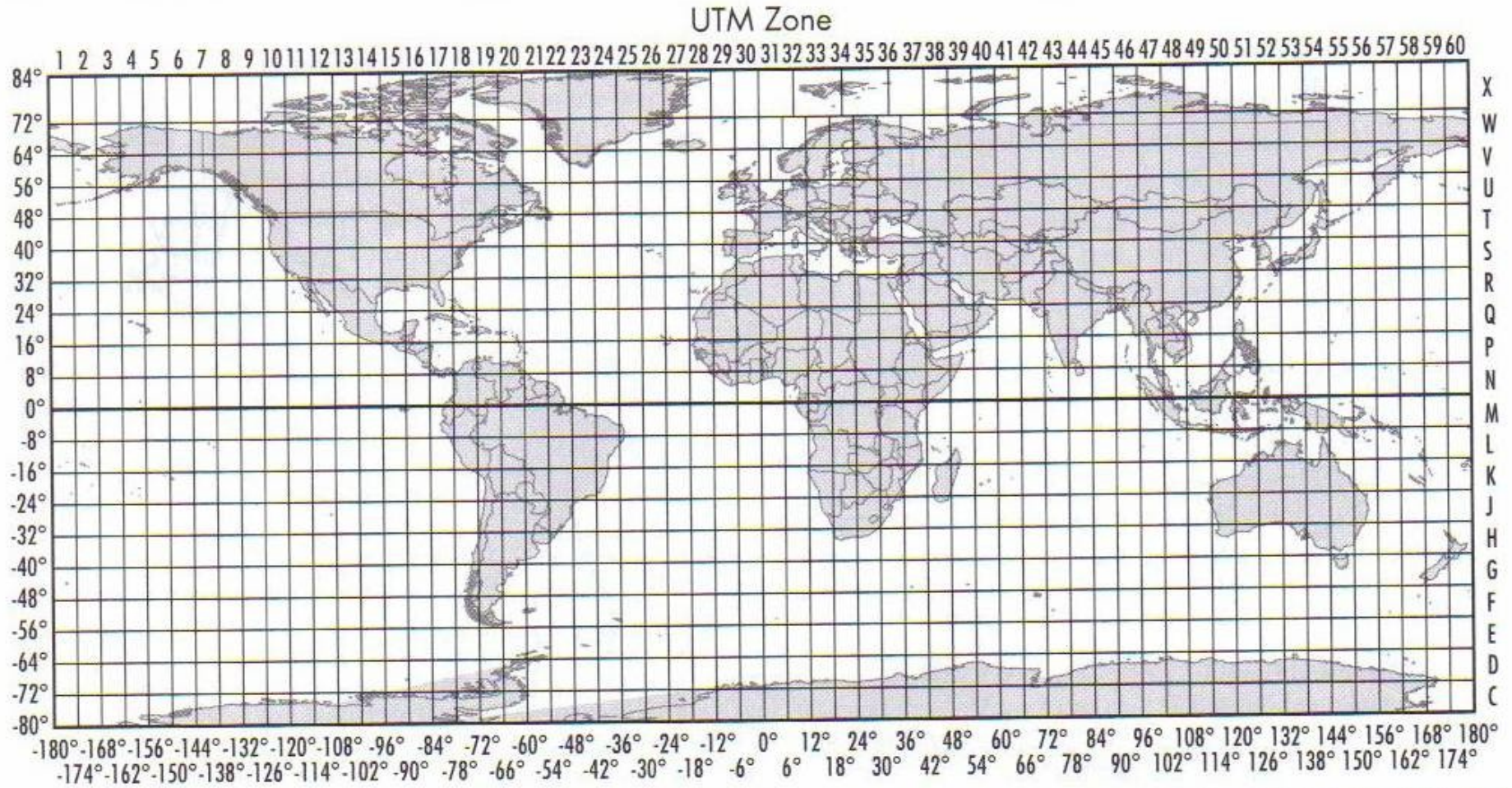
- **Meridians:** Equally spaced straight lines.
- **Parallels:** Unequally spaced straight lines, perpendicular to the meridians. Spacing increases toward the poles.
- Direction true between any two points and reasonably true shapes and distances within 15° of equator; High distortion for points near the poles

Universal Transverse Mercator (UTM) Coordinate System

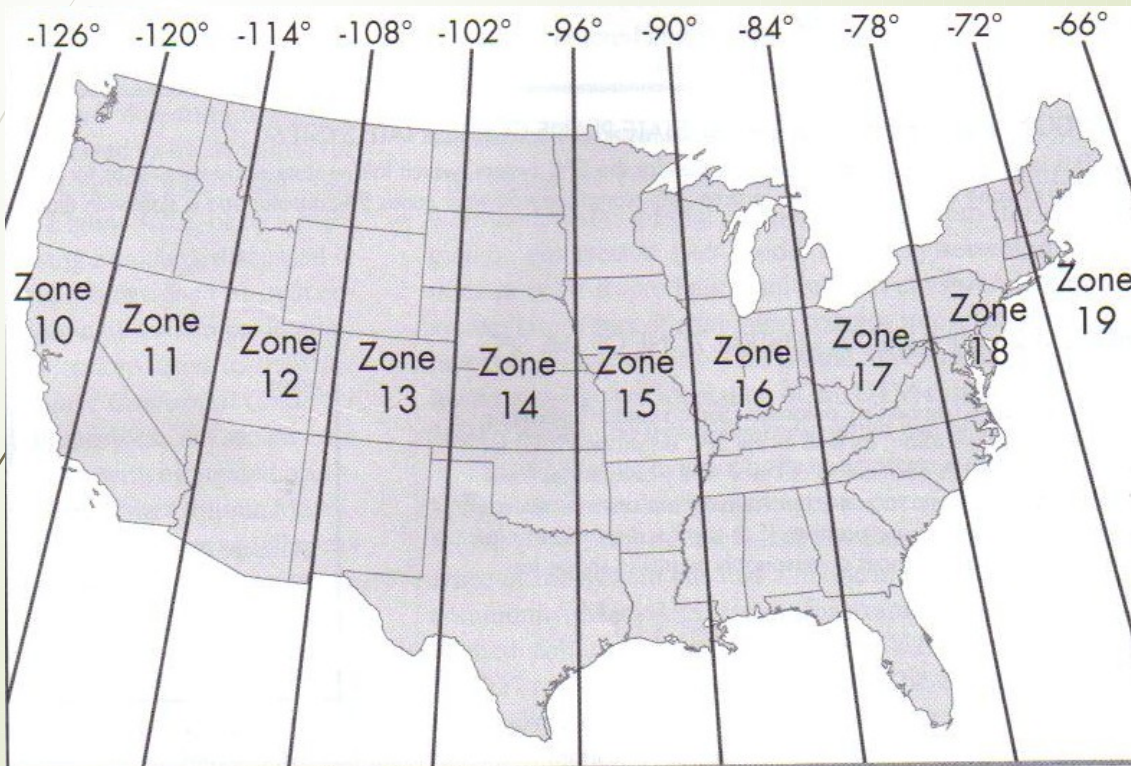
- **UTM is a projected coordinate system**, developed by several allied nations after World War II and is widely used in the world.
- **WGS84 ellipsoid** is used as the underlying model of the Earth in the UTM coordinate system.
- **The UTM coordinate system** is a grid-based method of specifying locations on the surface of the Earth.
- In the US, many spatial datasets or data layers are in UTM format (usually **NAD27 and NAD83 datum**).
- The UTM covering the world from 80°S to 84°N is not a single map projection. Instead, it is divided into **60 zones**. Each of 60 zones is based on a secant Transverse Mercator projection, 6° wide with a central meridian running at the center of the zone.
- The first zone starts at 180° and extends from 180° to 174°W as well as from 80°S to 84°N and its central meridian is at 177°W.

Universal Transverse Mercator (UTM) Coordinate System

Dent et al., 2009

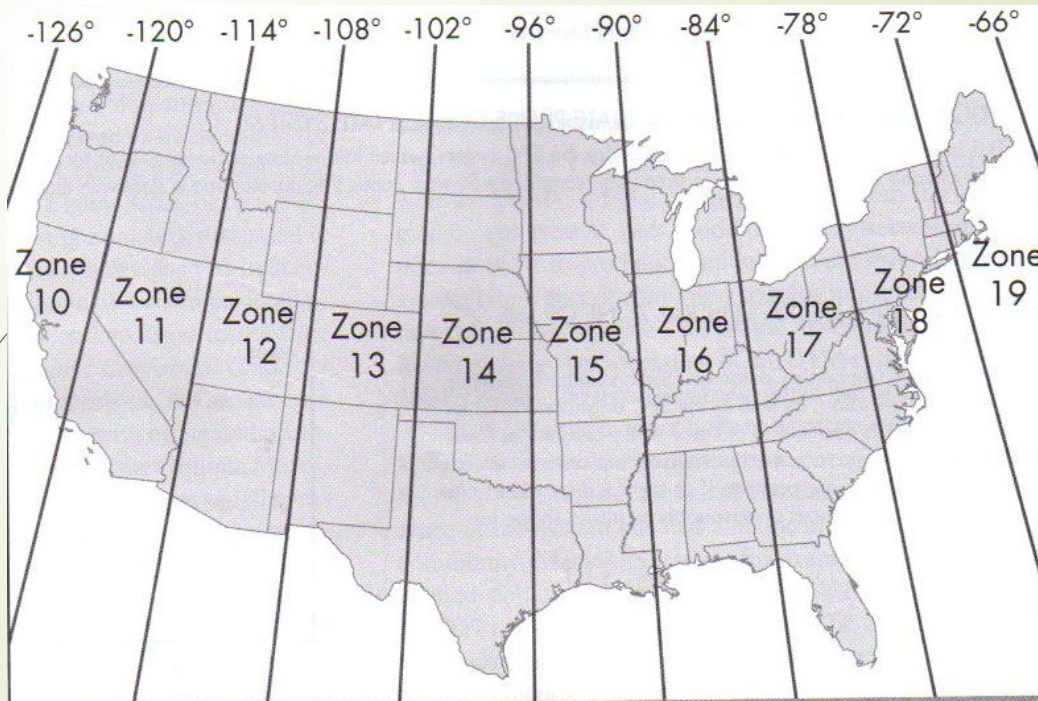


Universal Transverse Mercator (UTM) Coordinate System



- The continental US has 10 zones (zones 10-19) that extend from 126°W to 66°W.

Universal Transverse Mercator (UTM) Coordinate System



Dent et al., 2009

Houston is located within zone ??, covering ?? - ?? of longitude



UTM Coordinate System

Question:

Given Anchorage's coordinates (-149.9167° , 61.2024°), which UTM zone is it located within



Georeferencing and conversion

1. **Geocoding**: converting street addresses to coordinates for mapping;
2. **Converting** a coordinate system to another;
3. **Georeferencing** one map or image to a specific coordinate system, or Geo-registration

Georeferencing and conversion

Geocoding: converting street addresses to coordinates for mapping. **Address locators** use a specific process to find a location:

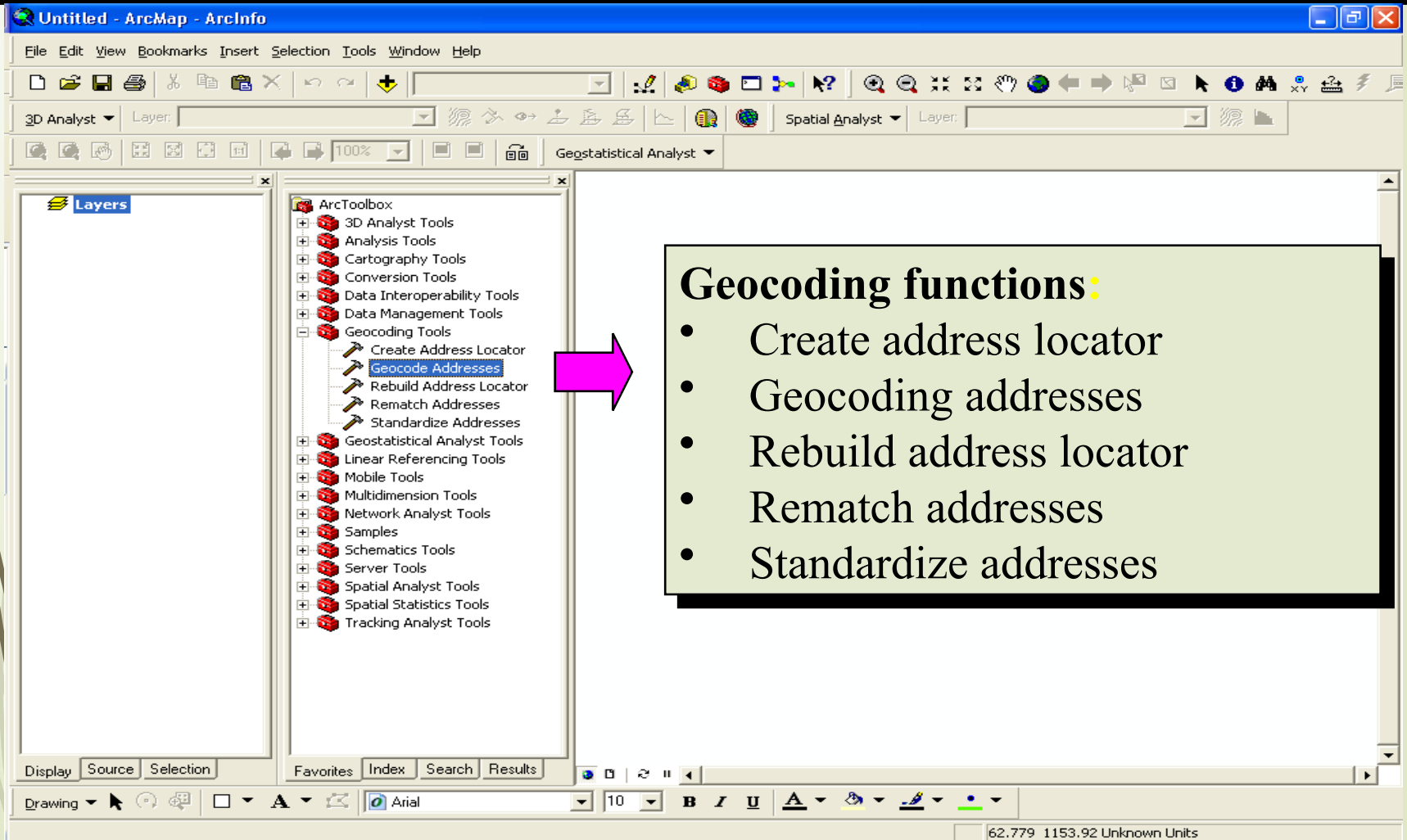
Address components

For example, the address “127 West Birmingham Drive, 92373” has five address components: **the street number, “127”; the street name, “Birmingham”; the street type, “Drive”; the street direction, “West”; and the postal code, “92373”.**

Each style of address locator parses an address into a different set of address components.

Georeferencing and conversion

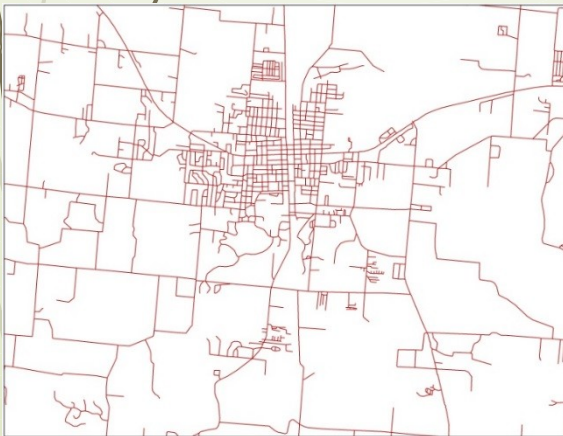
Geocoding: converting street addresses to coordinates for mapping



Georeferencing and conversion

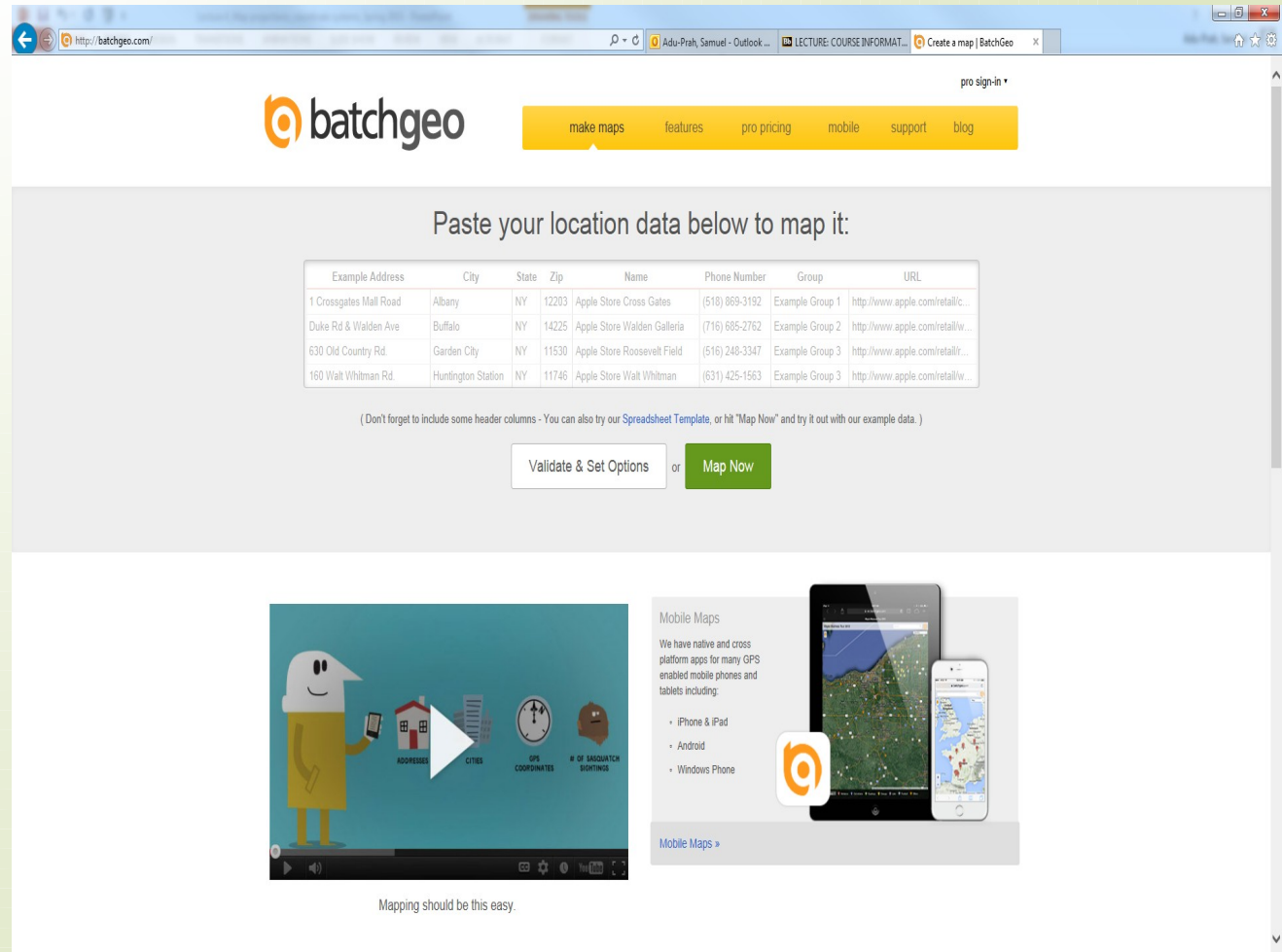
Geocoding: converting street addresses to coordinates for mapping

Street name	City	Zipcode
121 N Parrish Ln	Carbondale	62901
P O Box 40	Carbondale	62903
1245 East Grand Ave	Carbondale	62901
801 S Lewis Ln	Carbondale	62901
200 N Springer St	Carbondale	62901
Faner Hall 2179	Carbondale	62901

[illegible]

Georeferencing and conversion

Geocoding: converting street addresses to coordinates using batchgeo



The screenshot shows the batchgeo.com website. At the top is the batchgeo logo and a navigation bar with links: make maps, features, pro pricing, mobile, support, and blog. Below the navigation bar is a section titled "Paste your location data below to map it:". Under this title is a table with example data. The table has columns: Example Address, City, State, Zip, Name, Phone Number, Group, and URL. Below the table is a note: "(Don't forget to include some header columns - You can also try our Spreadsheet Template, or hit 'Map Now' and try it out with our example data.)". Below the note are two buttons: "Validate & Set Options" and "Map Now". At the bottom of the page, there is a section titled "Mobile Maps" with a video player on the left and text on the right. The video player shows a cartoon character holding a phone, with icons for addresses, cities, GPS coordinates, and a map. The text on the right says: "We have native and cross platform apps for many GPS enabled mobile phones and tablets including:" followed by a list: iPhone & iPad, Android, and Windows Phone. Below the list is a link "Mobile Maps >".

Example Address	City	State	Zip	Name	Phone Number	Group	URL
1 Crossgates Mall Road	Albany	NY	12203	Apple Store Cross Gates	(516) 869-3192	Example Group 1	http://www.apple.com/retail/c...
Duke Rd & Walden Ave	Buffalo	NY	14225	Apple Store Walden Galleria	(716) 685-2762	Example Group 2	http://www.apple.com/retail/w...
630 Old Country Rd.	Garden City	NY	11530	Apple Store Roosevelt Field	(516) 248-3347	Example Group 3	http://www.apple.com/retail/r...
160 Walt Whitman Rd.	Huntington Station	NY	11746	Apple Store Walt Whitman	(631) 425-1563	Example Group 3	http://www.apple.com/retail/w...

(Don't forget to include some header columns - You can also try our [Spreadsheet Template](#), or hit "Map Now" and try it out with our example data.)

[Validate & Set Options](#) or [Map Now](#)

Mobile Maps

We have native and cross platform apps for many GPS enabled mobile phones and tablets including:

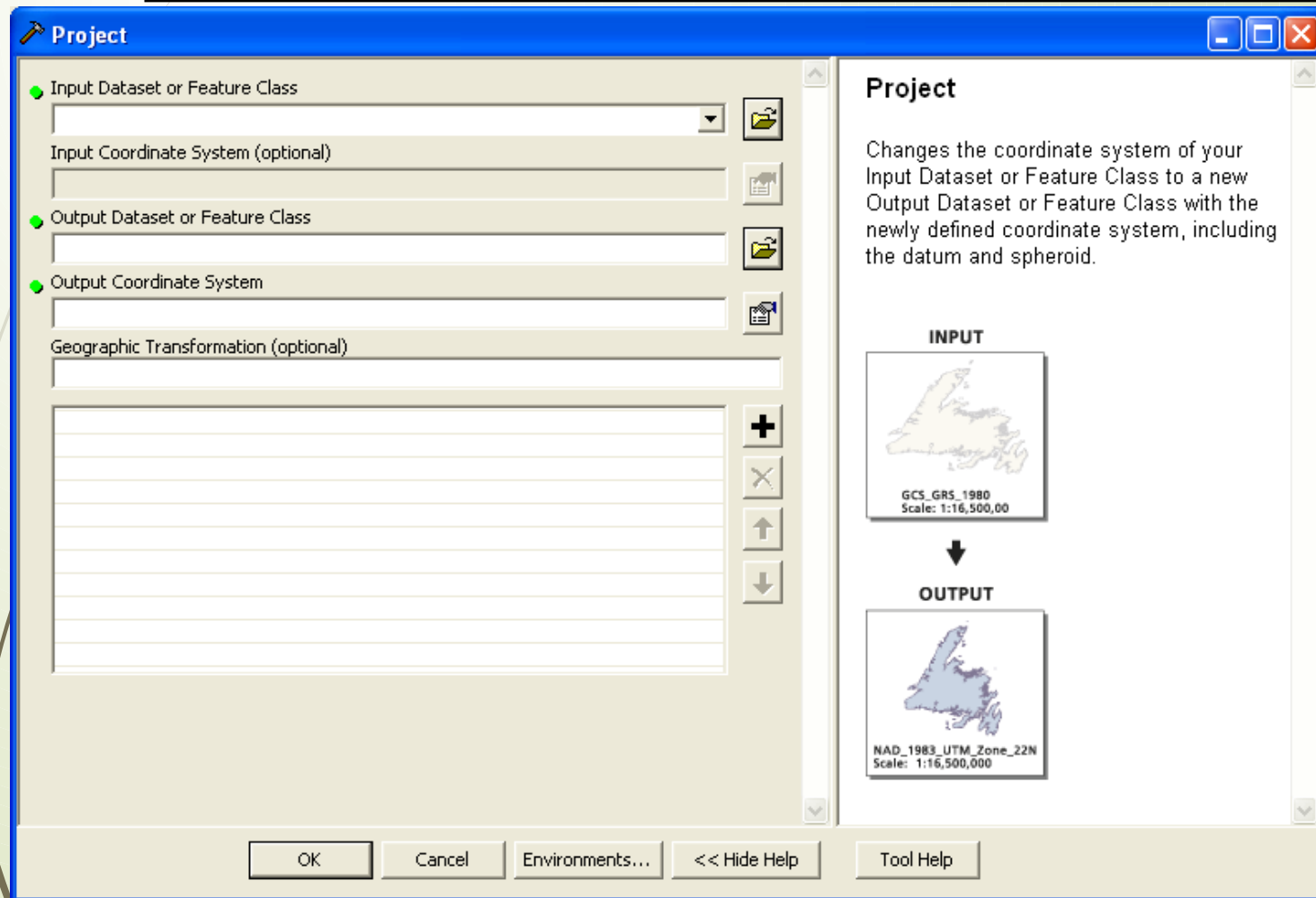
- iPhone & iPad
- Android
- Windows Phone

[Mobile Maps >](#)

<http://www.batchgeo.com/>

Georeferencing and conversion

Converting a coordinate system to another



Georeferencing and conversion

Georeferencing one map or image to a specific coordinate system, or
Geo-registration

a) Raster dataset to be georeferenced

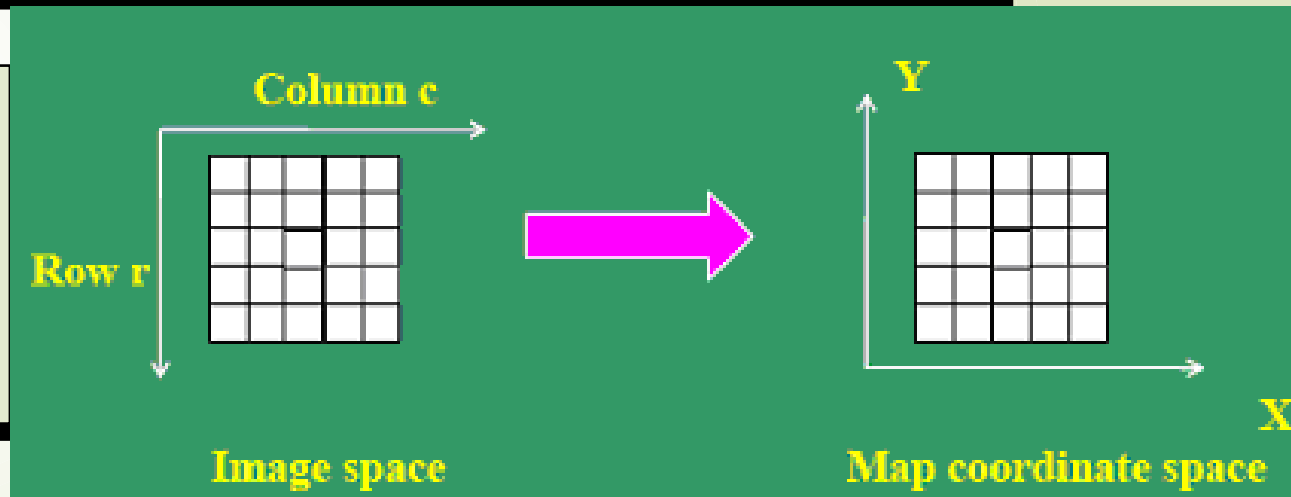


b) Raster dataset with map coordinate system



Georeferencing and conversion

Georeferencing one map or image to a specific coordinate system, or Geo-registration



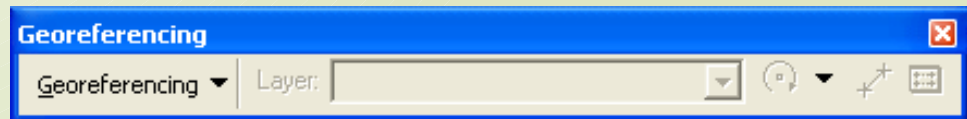
Link Table

Link	X Source	Y Source	X Map	Y Map	Residual
1	10.128302	2.632075	302822.603774	4165459.632075	3.11914
2	4.643277	8.895736	302062.358491	4166329.462264	2.10868
3	17.372302	15.374114	303815.716981	4167240.386792	0.71057
4	9.346338	13.226113	302713.018868	4166932.179245	4.46191
5	4.771656	19.230488	302082.905660	4167767.764151	2.32776
6	17.197934	11.395454	303802.018868	4166671.915094	0.72368
7	9.315212	16.195122	302706.169811	4167343.122642	2.82019
8	4.527135	13.340727	302055.509434	4166945.877358	4.07011
9	9.831124	7.085608	302774.660377	4166069.198113	6.13041

☒ Auto Adjust Transformation: 2nd Order Polynomial Total RMS Error: 3.37692

Load... Save... Restore From Dataset OK

$$\begin{cases} x_{map} = a + bx_{source} + cy_{source} \\ y_{map} = d + ex_{source} + fy_{source} \end{cases}$$



GPS Project Planning



1. Define objectives of the survey

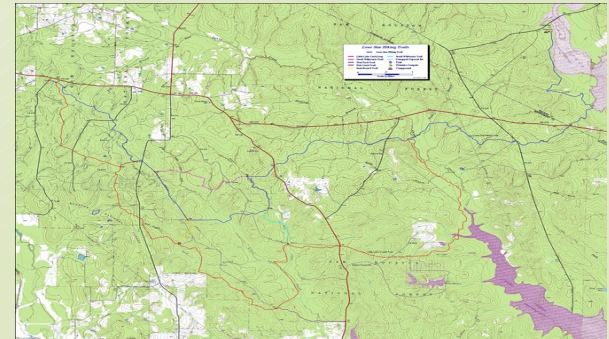
- ▢ This determines
 - ▢ The level of accuracy required
 - ▢ Type of equipment
 - ▢ Level of preparation



2. Define project area

This step is designed to establish the overall project area and define the limits of the survey.

- Maps and/or aerial photos should be utilized extensively to familiarize the crew with the area prior to the actual field work.
- For identifying the study area and surrounding environment, 7.5-minute topographic maps are ideal.
- For locating particular sites by address, a local street map will be required.
- A complete understanding of the transportation network in the project area



3. Determine observation window and schedule operations

□ **Mission planning**

- Involves determining the precise window of satellite availability and scheduling accordingly



GPS TECHNIQUES

- Four GPS survey and processing techniques
 - Autonomous GPS
 - Real-Time Differential GPS (RTD)
 - Real-Time Kinematic GPS (RTK)
 - Post-Processed Differential (PPD)

LEVEL OF ACCURACY

- Before choosing a GPS technique, the accuracy needs of the project need to be analyzed.
- Factors that are considered during this analysis include:
 - What is the purpose of the resulting data?
 - Will the data be used for multiple purposes?
 - What are the horizontal and vertical requirements of the project?

Why use GPS to collect locations

- Permits linking of additional data to geographic links
 - E.g. Census data, facility data, other geo-located data
- Permits geographic grouping of data beyond region
 - E.g. Climactic zones, agricultural zones, etc.
- Permits mapping

GPS Data Capture- -Insitu Measurement

Global Positioning System (GPS) Data Collection



a. Data collection using a hand-held receiver.



b. Two hand-held GPS receivers.



c. A GPS receiver installed on a tractor used for precision agriculture.

GPS Collection Process

1. The GPS collection teams collect one waypoint (location) for each assigned location
2. The waypoints are captured as close to the geographic center of the location as possible
3. Saved information are waypoint name (number), latitude, longitude, and altitude to the GPS unit
4. These information should also be written down on GPS form that will serve as a paper back-up if any digital data in the GPS unit is lost or corrupted



GPS Collection Teams

1. GPS data collection may be incorporated into different phases of survey:

- If the GPS units are used to update the sample frame, then listing teams are responsible for collecting the GPS data
- GPS data collection may also take place during interviews, in which case one person on a team may be responsible for collecting the GPS waypoints



How to use GPS

1. Finding a “Good” Location to Collect a GPS Waypoint
2. Activate GPS
3. Identify and mark location (“waypoint”)
4. Record location on form

Post Field GPS data checking and cleaning

- When GPS data return from field work, the GPS Data Collection Coordinator checks that the GPS data are correct
- Use ***GPS TrackMaker***, a free geographic information system, to plot the GPS points and check for common errors

Common errors in the GPS data

- Duplicate or Paired Points
- Missing Data
- Administrative Mismatches
- Hand-Entry Typos